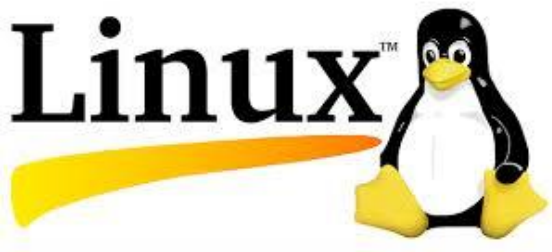
Linux and Windows Server Operating System Fundamentals



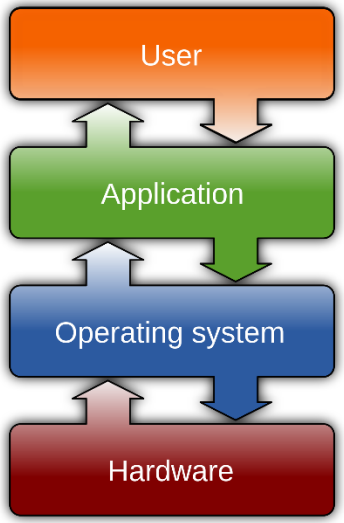
By,

Mr. Kumar Pudashine, (MEng, AIT, Bangkok)
PMP, CISA, CISM, CRISC, CNDA, CDCP, COBIT 5, CCNP (Enterprise), JNCIA, CEH v9, ITIL, ISO 27001:2022, AcitivIdentity Certified
Founder and President, Global Cybersecurity Community Forum
IEEE, ISACA, PMI Member

May 2025

Operating System: Definition ??

An operating system (OS) is a fundamental software component that manages computer hardware and software resources, providing essential services for computer programs. It serves as a bridge between users and the computer hardware, ensuring the efficient and secure operation of the entire system.



Operating System: Definition ??

Operating System as an Extended Machine

Operating System as a Resource Manager

Operating System: OS as an Extended Machine

- **Abstraction**: The OS provides an abstraction of the underlying hardware, presenting a more user-friendly interface and simplifying the complexity of hardware operations.

- **User Interface**: It offers a higher-level interface for users and applications, making it easier to perform tasks without needing to understand the intricate details of the hardware.

- **System Calls and APIs**: The OS supplies system calls and application programming interfaces (APIs) that allow software developers to interact with the hardware in a standardized way, enhancing portability and ease of development.

- **Virtual Machine**: By abstracting hardware resources, the OS can create the illusion of a virtual machine that supports multiple applications running concurrently as if each had its dedicated resources.

Operating System: OS as a Resource Manager

- **Process Management**: The OS allocates CPU time to various processes, schedules their execution, and handles multitasking to ensure efficient use of the processor.
- **Memory Management**: It manages the system's memory, assigning space to processes, handling memory allocation and deallocation, and optimizing the use of RAM to prevent conflicts and maximize performance.
- **Storage Management**: The OS controls data storage on devices, organizing files and directories, managing disk space, and providing access to files in a secure and efficient manner.
- **Device Management**: It oversees the operation of hardware devices, ensuring that input/output operations are performed correctly and efficiently, and manages device drivers to facilitate communication between the hardware and software.
- **Network Management**: The OS manages network connections, handles data transmission and reception, and ensures reliable communication between networked devices.

Linux Distributions

Linux distributions, often referred to as "distros," are variations of the Linux operating system that package the Linux kernel with a set of software applications, utilities, and other components, tailored to meet specific user needs and preferences.

Key Components : Linux Distributions

1. **Linux Kernel**: The core of the operating system, managing hardware resources and system operations.
2. **GNU Utilities and Tools**: Basic tools for file manipulation, text processing, and system management.
3. **Package Manager**: Software that automates the installation, updating, and removal of software packages (e.g., APT, YUM, Pacman).
4. **Desktop Environment**: Graphical user interface components (e.g., GNOME, KDE, Xfce).
5. **Applications**: Pre-installed software such as web browsers, office suites, media players, and more.
6. **Configuration Tools**: Utilities for system configuration and management.

Key Components : Linux Distributions

Ubuntu

- **Target Audience**: Beginners and general users.
- **Features**: User-friendly, strong community support, regular updates.
- **Desktop Environments**: GNOME (default), KDE (Kubuntu), Xfce (Xubuntu), LXQt (Lubuntu).

Fedora

- **Target Audience**: Developers, enthusiasts, and those who want the latest technology.
- **Features**: Cutting-edge software, strong focus on free and open-source software.
- **Desktop Environments**: GNOME (default), KDE, Xfce, and others.

Key Components : Linux Distributions

Debian

- **Target Audience**: Advanced users and servers.
- **Features**: Stability, large repository of software packages, strong community governance.
- **Desktop Environments**: GNOME, KDE, Xfce, LXDE, and more.

Arch Linux

- **Target Audience**: Advanced users who prefer a DIY (Do It Yourself) approach.
- **Features**: Rolling release model, high customizability, minimalistic base system.
- **Desktop Environments**: User's choice; highly customizable.

Key Components : Linux Distributions

Kali Linux

- **Target Audience**: Security professionals and ethical hackers.
- **Features**: Pre-installed security and penetration testing tools.
- **Desktop Environments**: Xfce (default), GNOME, KDE, and others.

Raspberry Pi OS (formerly Raspbian)

- **Target Audience**: Raspberry Pi users.
- **Features**: Optimized for the Raspberry Pi hardware, educational tools, and lightweight design.
- **Desktop Environments**: PIXEL (default, based on LXDE).

Linux : File System ??

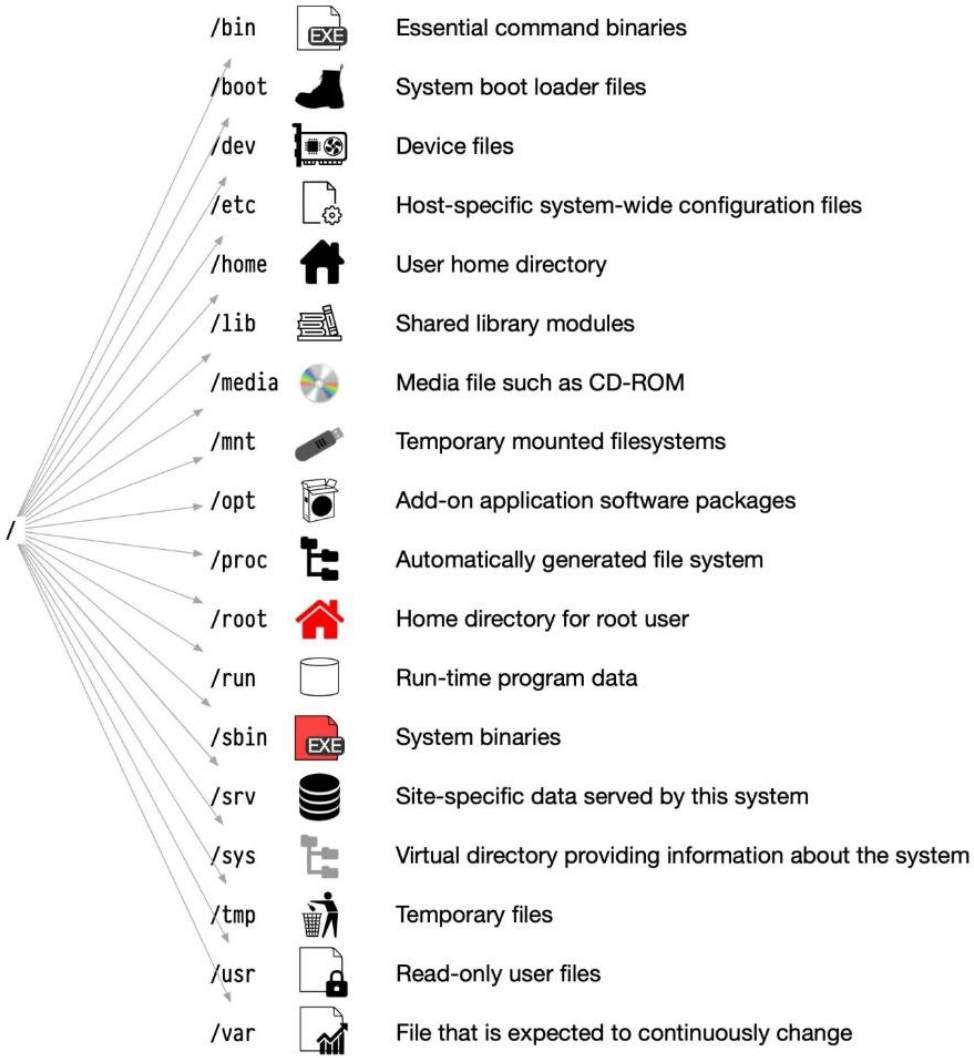
A file system in Linux is a method and data structure that the operating system uses to control how data is stored and retrieved on a storage device. Without a file system, data placed in a storage medium would be one large body of data with no way to tell where one piece of data stops and the next begins.

Linux : Key Components of a File System

Key Components

- **Inodes**
  - Each file and directory is represented by an inode (index node).
  - An inode contains metadata about the file such as its size, ownership, permissions, and timestamps.
  - The inode does not contain the file name or the actual data, only metadata and pointers to the data blocks.
- **Directories**
  - Directories are special types of files that contain mappings between file names and inode numbers.
  - They help organize files in a hierarchical structure.
- **Data Blocks**
  - Data blocks are the actual locations on the disk where file data is stored.
  - The inode points to these blocks.

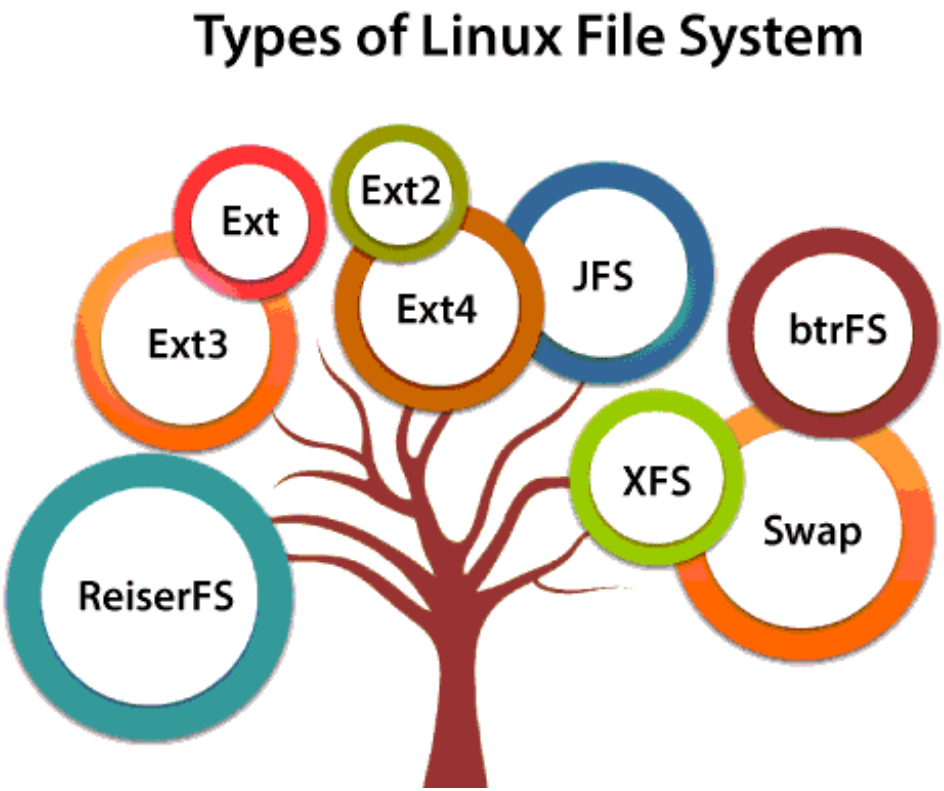
Linux : File System Hierarchy



Linux : File System Hierarchy

/(Root): The top-level directory in the hierarchy.
/bin: Essential command binaries (executables) needed for system booting.
/boot: Static files of the boot loader, including the kernel.
/dev: Device files representing hardware devices.
/etc: System-wide configuration files and scripts.
/home: User home directories, **e.g., /home/username.**
/lib: Essential shared libraries and kernel modules.
/media: Mount points for removable media such as USB drives.
/mnt: Temporary mount point for the system administrator.
/opt: Add-on application software packages.
/proc: Virtual filesystem providing process and system information.
/root: Home directory for the root user.
/sbin: Essential system binaries.
/srv: Data for services provided by the system.
/tmp: Temporary files.

Types of Linux File System



Types of Linux File System

1. Ext, Ext2, Ext3 and Ext4 file system

The file system Ext stands for **Extended File System**. It was primarily developed for **MINIX OS**. The Ext file system is an older version, and is no longer used due to some limitations.

Ext2 is the first Linux file system that allows managing two terabytes of data. Ext3 is developed through Ext2; it is an upgraded version of Ext2 and contains backward compatibility. The major drawback of Ext3 is that it does not support servers because this file system does not support file recovery and disk snapshot.

2. JFS File System

JFS stands for **Journaled File System**, and it is developed by **IBM for AIX Unix**. It is an alternative to the Ext file system. It can also be used in place of Ext4, where stability is needed with few resources. It is a handy file system when CPU power is limited.

Types of Linux File System

3. ReiserFS File System

ReiserFS is an alternative to the Ext3 file system. It has improved performance and advanced features. In the earlier time, the ReiserFS was used as the default file system in SUSE Linux, but later it has changed some policies, so SUSE returned to Ext3.

4. XFS File System

XFS file system was considered as high-speed JFS, which is developed for parallel I/O processing. NASA still using this file system with its high storage server (300+ Terabyte server).

5. Btrfs File System

Btrfs stands for the **B tree file system**. It is used for fault tolerance, repair system, extensive storage configuration, and more.

Basic Linux Commands

File and Directory Commands

1. `ls` - List directory contents.

```bash
ls              # List files in the current directory
ls -l           # Long listing format
ls -a           # Include hidden files
ls -lh          # Long listing with human-readable file sizes
```

2. `cd` - Change the current directory.

```bash
cd /path/to/directory    # Change to specified directory
cd ~                     # Change to home directory
cd ..                    # Move up one directory level
```

Basic Linux Commands

3. `pwd` - Print working directory; shows the current directory path.

```bash
pwd
```

4. `mkdir` - Create a new directory.

```bash
mkdir new_directory
```

5. `rmdir` - Remove an empty directory.

```bash
rmdir empty_directory
```

Basic Linux Commands

6. `cp` - Copy files or directories.

```bash
cp source_file destination_file              # Copy a file
cp -r source_directory destination_directory  # Copy a directory recursively
```

7. `mv` - Move or rename files or directories.

```bash
mv old_name new_name            # Rename a file or directory
mv file.txt /path/to/destination/  # Move a file to another directory
```

8. `rm` - Remove files or directories.

```bash
rm file.txt             # Remove a file
rm -r directory         # Remove a directory and its contents recursively
```

Basic Linux Commands

File Viewing and Editing Commands

9. `cat` - Concatenate and display file contents.

```bash
cat file.txt
```

10. `less` - View file contents one screen at a time.

```bash
less file.txt
```

11. `head` - Display the first part of a file.

```bash
head file.txt
head -n 20 file.txt  # Display the first 20 lines
```

Basic Linux Commands

12. `**tail**` - Display the last part of a file.

bash Copy code

```bash
tail file.txt
tail -n 20 file.txt  # Display the last 20 lines
tail -f file.txt     # Follow the file as it grows
```

System Information

15. `df` - Display disk space usage.

```bash
df -h  # Human-readable format
```

16. `du` - Display disk usage of files and directories.

```bash
du -h /path/to/directory  # Human-readable format
```

17. `free` - Display memory usage.

```bash
free -h  # Human-readable format
```

18. `uname` - Display system information.

```bash
uname -a  # All system informati
```

Basic Linux Commands

Process Management

19. `ps` - Display currently running processes.

```bash
ps aux  # Detailed process list
```

20. `top` - Display real-time system resource usage.

```bash
top
```

21. `kill` - Terminate a process by PID.

```bash
kill PID          # Terminate process with specified PID
kill -9 PID       # Forcefully terminate a process
```

Basic Linux Commands

Network Commands

22. `ping` - Check network connectivity.

```bash
ping example.com
```

23. `ifconfig` / `ip` - Display or configure network interfaces.

```bash
ifconfig            # Display network interfaces (older systems)
ip addr show        # Display network interfaces (modern systems)
```

24. `netstat` - Display network connections, routing tables, and more.

```bash
netstat -tuln  # List listening ports
```

Basic Linux Commands

Viewing Permissions and Ownership

Use the `ls -l` command to view the detailed list of files with their permissions and ownership.

bash Copy code

```bash
ls -l
```

Example output:

csharp Copy code

```csharp
-rwxr-xr-- 1 user group 4096 Jan 1 12:00 file.txt
```

- `-rwxr-xr--`: File type and permissions
- `1`: Number of hard links
- `user`: Owner
- `group`: Group
- `4096`: File size in bytes
- `Jan 1 12:00`: Last modification date and time
- `file.txt`: File name

Basic Linux Commands

Adding a New User

To add a new user in Linux, you can use the `useradd` command. Here's a basic example:

1. **Create a New User:**

   ```sh
   sudo useradd username
   ```

   This command creates a new user with the specified username.

2. **Create a Home Directory:**

   By default, `useradd` may not create a home directory. To ensure a home directory is created, use the `-m` option:

   ```sh
   sudo useradd -m username
   ```

Basic Linux Commands

3. **Set a Password for the New User:**

   After creating the user, set a password using the `passwd` command:

   ```sh
   sudo passwd username
   ```

4. **Add Additional User Details:**

   You can add additional information such as a comment (full name), user's home directory, default shell, etc., using various options with the `useradd` command. For example:

   ```sh
   sudo useradd -m -c "Full Name" -s /bin/bash username
   ```

   - `-c "Full Name"` sets the full name (comment).
   - `-s /bin/bash` sets the default shell to `/bin/bash`.

Basic Linux Commands

Adding a New Group

To add a new group, use the `groupadd` command:

1. **Create a New Group:**

```sh
sudo groupadd groupname
```

   This command creates a new group with the specified group name.

Adding a User to a Group

To add a user to a group, you can use the `usermod` command with the `-aG` options:

1. **Add a User to a Group:**

```sh
sudo usermod -aG groupname username
```

   - `-a` (append) ensures the user is added to the new group without being removed from other groups.
   - `-G groupname` specifies the group to which the user should be added.

Basic Linux Commands

Examples

1. Create a New User with a Home Directory and Bash Shell:

```sh
sudo useradd -m -s /bin/bash johndoe
sudo passwd johndoe
```

2. Create a New Group:

```sh
sudo groupadd developers
```

3. Add an Existing User to a New Group:

```sh
sudo usermod -aG developers johndoe
```

Linux File Permissions: Introduction

- ✓ Linux file permissions and security are fundamental aspects of the operating system, providing control over who can read, write, and execute files.
- ✓ Understanding and managing these permissions are crucial for maintaining a secure and well-functioning system.

Linux File Permissions: Basics

File Permissions

In Linux, each file and directory has three types of permissions for three categories of users:

1. **User (u)**: The owner of the file.
2. **Group (g)**: Users who are members of the file's group.
3. **Others (o)**: All other users.

Permissions are represented in a triplet format (rwx):

- **r** (read): Permission to read the file or list directory contents.
- **w** (write): Permission to modify the file or directory.
- **x** (execute): Permission to execute the file (if it's a script or binary) or access the directory.

Viewing Permissions

Use the `` `ls -l` `` command to view permissions of files and directories. The output looks like this:

diff Copy code

```diff
-rwxr-xr--
```

Linux File Permissions: Viewing Permissions

Viewing Permissions

Use the `ls -l` command to view permissions of files and directories. The output looks like this:

diff Copy code

```diff
-rwxr-xr--
```

This breaks down as:

- `-` (file type): `-` for a regular file, `d` for a directory, `l` for a symbolic link, etc.
- `rwx` (user permissions): Read, write, and execute for the file owner.
- `r-x` (group permissions): Read and execute for the group.
- `r--` (other permissions): Read-only for others.

Linux File Permissions: Changing Permissions

Changing Permissions

Use the `chmod` command to change permissions. There are two methods: symbolic and numeric.

Symbolic Method:

- `u`, `g`, `o` for user, group, others.
- `+`, `-`, `=` to add, remove, or set permissions.

Example:

bash Copy code

```bash
chmod u+x myfile    # Add execute permission to the owner
chmod g-w myfile    # Remove write permission from the group
chmod o=r myfile    # Set read-only permission for others
```

Numeric Method:

- Permissions are represented by numbers: `r=4`, `w=2`, `x=1`.
- Combine them to set permissions.

Linux File Permissions: Changing Permissions

Example:

bash Copy code

```bash
chmod 755 myfile    # rwx for owner, r-x for group and others
chmod 644 myfile    # rw- for owner, r-- for group and others
```

Linux File Ownership: chown

1. **Change the Owner of a File:**

```sh
sudo chown newowner filename
```

   This command changes the owner of the file to `newowner`.

2. **Change the Group of a File:**

```sh
sudo chown :newgroup filename
```

   This command changes the group of the file to `newgroup`.

3. **Change Both Owner and Group of a File:**

```sh
sudo chown newowner:newgroup filename
```

Linux File Ownership: chown

Examples

1. Change the Owner of a File to `john`:

```sh
sudo chown john example.txt
```

2. Change the Group of a File to `developers`:

```sh
sudo chown :developers example.txt
```

3. Change the Owner to `john` and the Group to `developers`:

```sh
sudo chown john:developers example.txt
```

Linux File Ownership: Recursive Change

Recursive Change

To change the ownership of a directory and all its contents recursively, use the `-R` option:

1. **Change the Owner of a Directory and All Its Contents:**

```sh
sudo chown -R newowner directoryname
```

2. **Change Both Owner and Group of a Directory and All Its Contents:**

```sh
sudo chown -R newowner:newgroup directoryname
```

Windows Server Operating System

Your trusted operating system for 30 years

Windows Server is the platform for building an infrastructure of connected applications, networks, and web services. As a Windows Server administrator, you've helped achieve your business' goals keeping the infrastructure secure, available, and flexible. Windows Server has been the foundation of Microsoft's ecosystem and continues to power the hybrid cloud network today.

Source
https://www.microsoft.com/en-us/windows-server

Windows Server Operating System: Common Versions

1.Windows Server 2012 and 2012 R2

- ✓ Introduced features like Hyper-V 3.0, improved PowerShell support, and Storage Spaces.

1.Windows Server 2016

- ✓ Added support for Docker containers, Nano Server, and enhanced security features like Shielded VMs and Just Enough Administration (JEA).

1.Windows Server 2019

- ✓ Introduced improvements in hybrid cloud integration, security enhancements, and Kubernetes support.

1.Windows Server 2022

- ✓ Enhanced security features with Secured-core server, better integration with Azure, and improvements in performance and scalability.

Roles and Features in Windows Server

Active Directory Domain Services: Provides directory services.

DHCP Server: Allocates IP addresses dynamically.

DNS Server: Resolves hostnames to IP addresses.

File and Storage Services: Manages file sharing and storage.

Web Server (IIS): Hosts websites and web applications.

Active Directory and Domain Controllers

Active Directory and Domain Controllers are two essential components that work together to provide secure user authentication and resource access on a Windows Server Network.

Analogy

Think of AD as a library catalog (centralized database) that stores information about all the books (users, groups, computers) in the library. Domain controllers are like librarians (servers) who verify your ID (authentication) and tell you which books you can borrow (authorization) based on your membership (group policy).

Active Directory: Key Components

1. **Domain:** A domain is a logical grouping of network objects (such as users, computers, and devices) that share the same Active Directory database.

2. **Forest:** A forest is a collection of one or more Active Directory domains that share a common schema but are configured to operate independently.

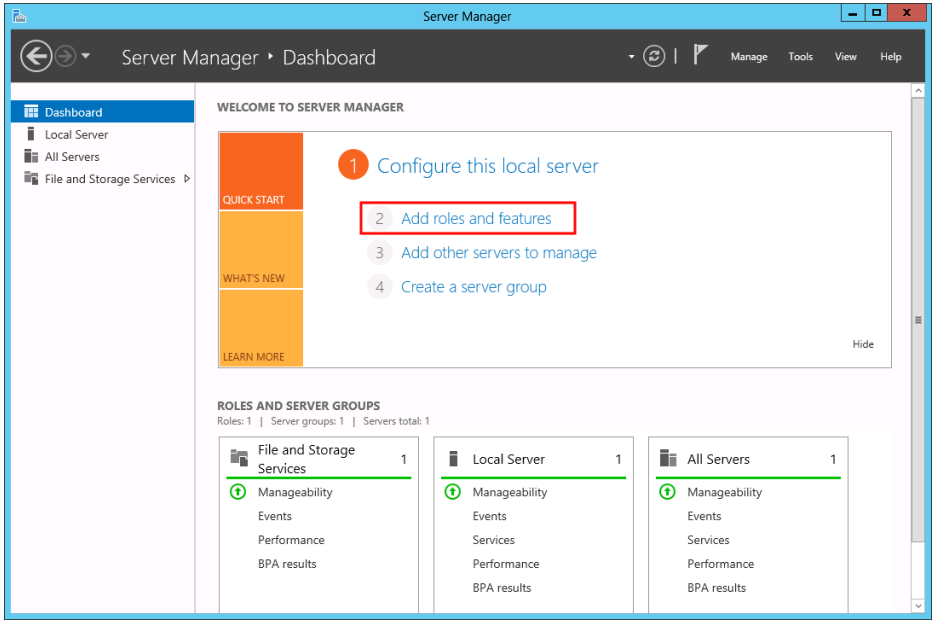
3. **Organizational Units (OUs):** These are containers within a domain that can hold users, groups, computers, and other OUs. They help organize the directory into manageable sections.

4. **Trusts:** Trust relationships between domains allow users in one domain to access resources in another domain.

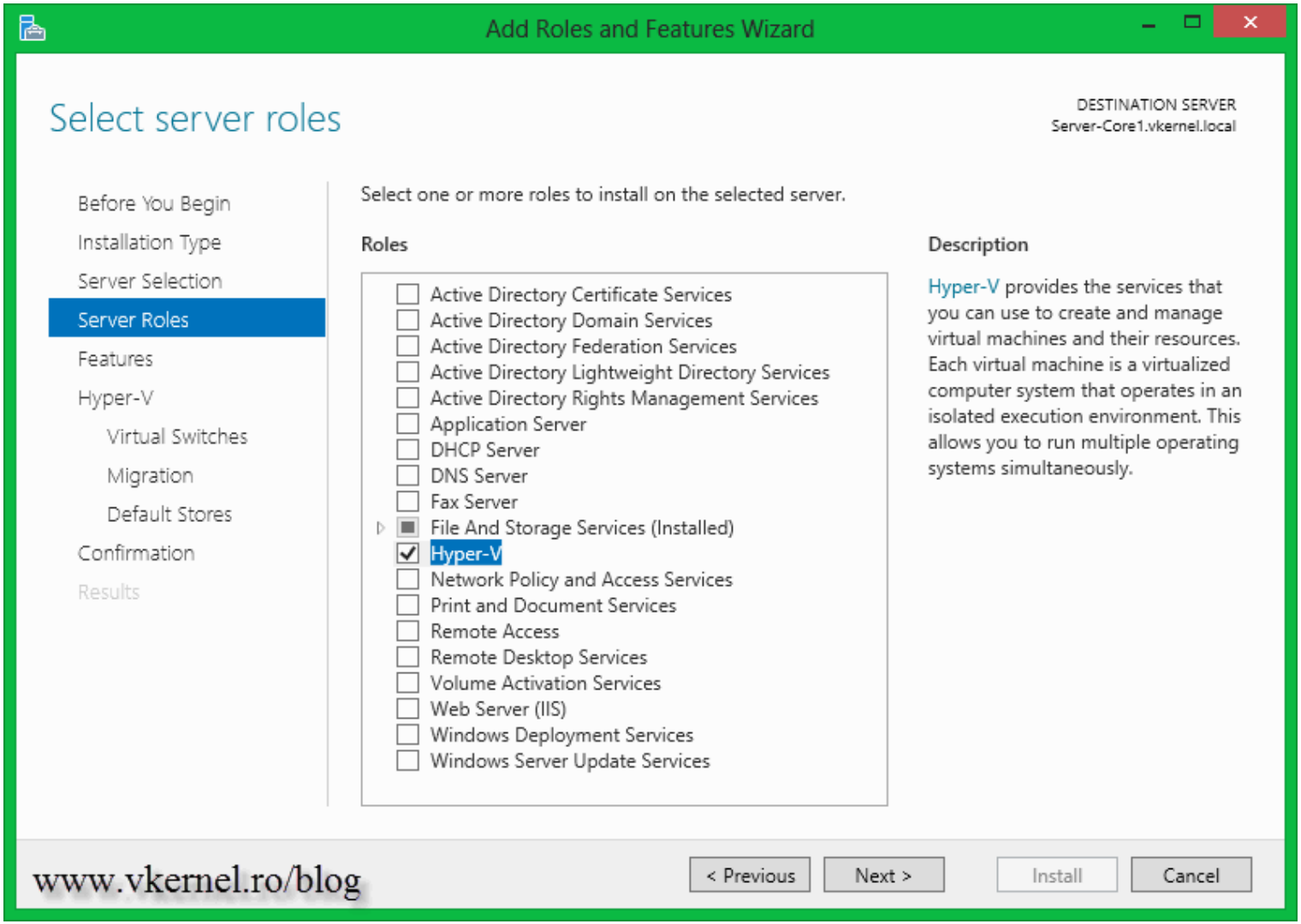
5. **Group Policy:** A feature that allows administrators to manage the configuration of operating systems, applications, and users' settings in an Active Directory environment.

Windows Server Roles and Features

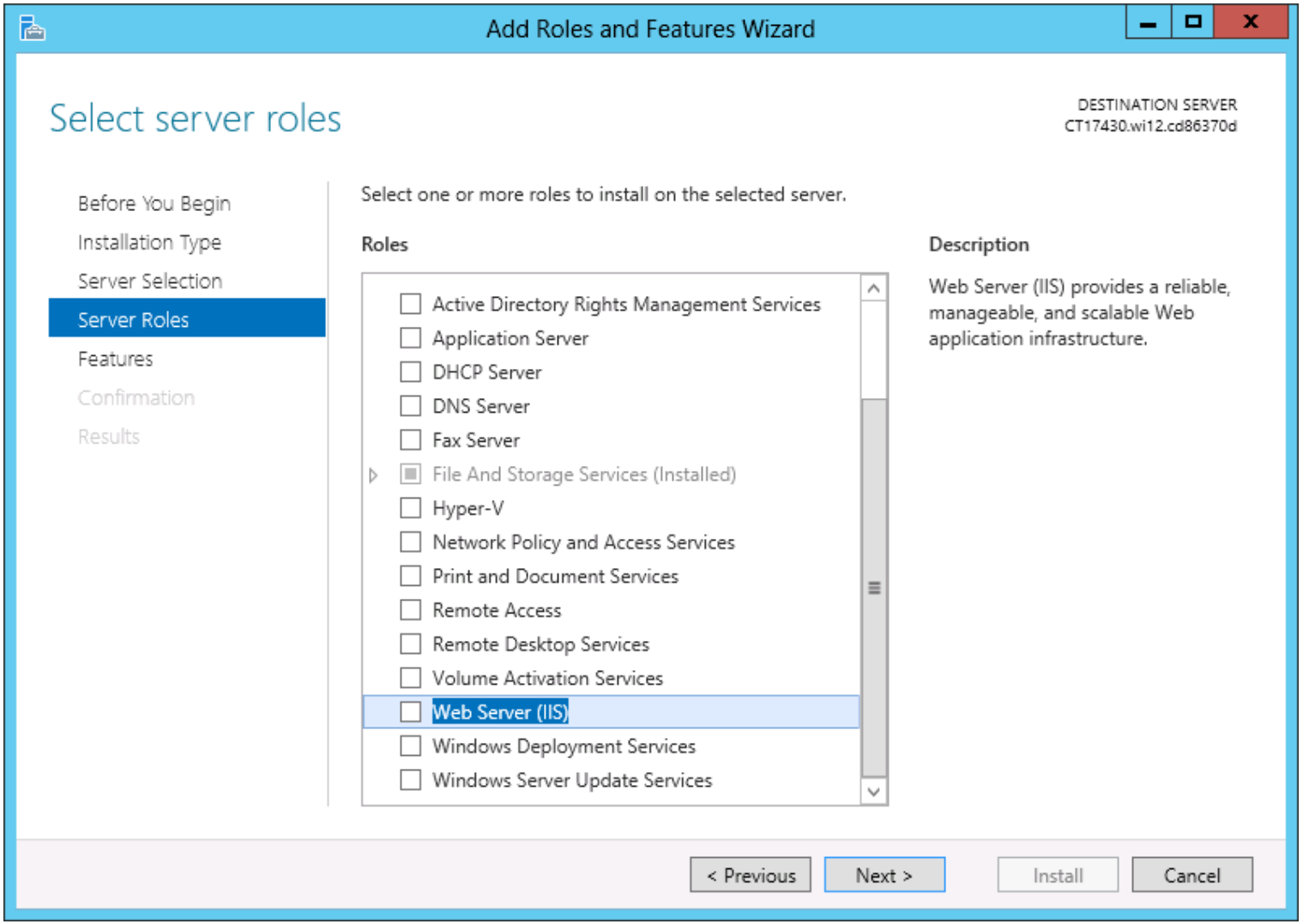
Windows Server Roles and Features are sets of functionalities that can be installed and configured on a Windows Server to provide specific services and capabilities. Understanding these roles and features is crucial for efficiently managing and deploying various services in a Windows Server environment.



Windows Server Roles and Features

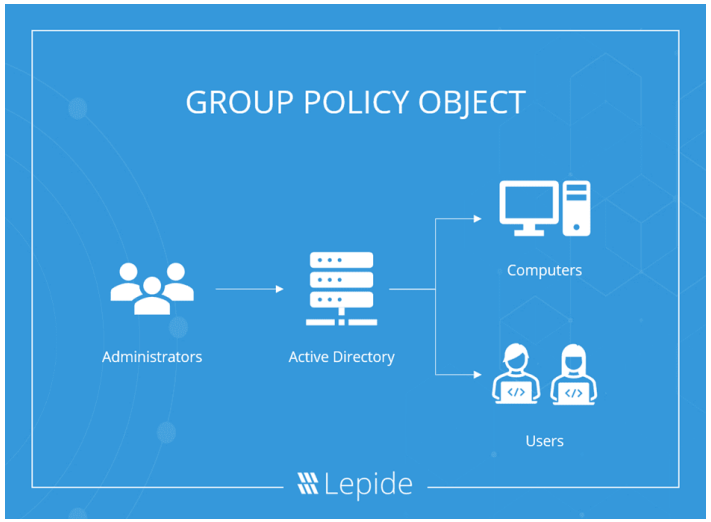


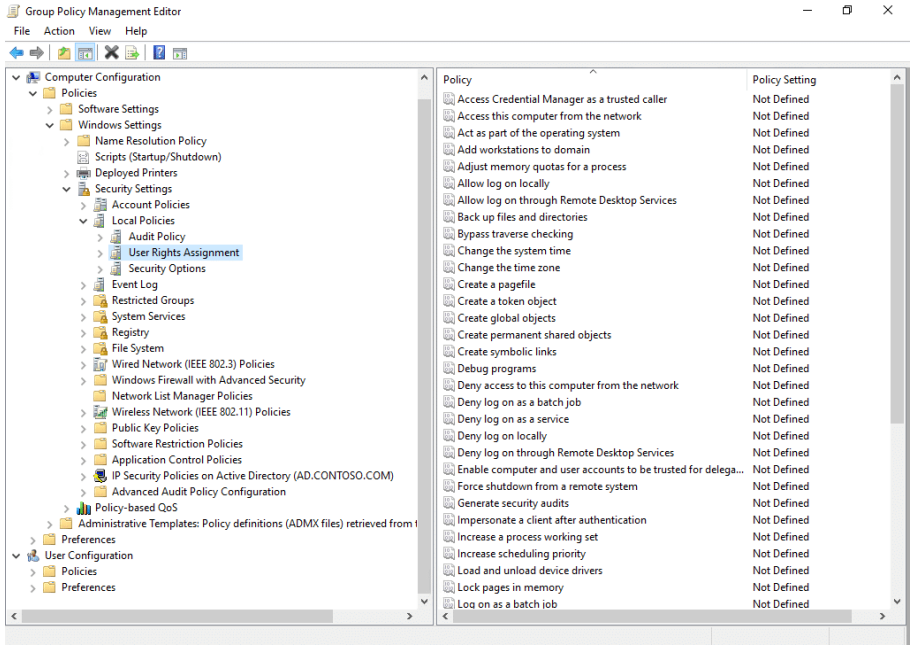
Windows Server Roles and Features



Windows Server : Group Policy

Group Policy is a feature of Windows Server that provides centralized management and configuration of operating systems, applications, and user settings in an Active Directory environment.





Windows Server : Group Policy

Group Policy Components

1. **Group Policy Objects (GPOs):**
   - GPOs are collections of settings created using the Group Policy Management Console (GPMC) that are applied to users and computers within an Active Directory domain.
2. **Group Policy Management Console (GPMC):**
   - A management tool that provides a single interface for managing GPOs across a domain.
3. **Group Policy Preferences:**
   - A set of extensions that provide advanced configuration settings beyond standard policies.
4. **Local Group Policy:**
   - Policies that are applied to a single computer and do not require Active Directory.

Linux & Windows Server LAB Exercise

1. Setup EC2 Amazon Linux Server
2. Use Basic Linux Commands and File Permissions
3. Setup EC2 Windows Server 2019/2022 from AWS Console
4. Setup Active Directory Domain Controller Server
5. Setup Domain Name Server
6. Setup Web Server

Thank You

If You have any Queries write to Me

@

Kumar@ncit.edu.np